

SALAMANCA AP, Spain

WMO: 082020

Lat: 40.9596N

Lon: 5.4982W

Elev: 794

StdP: 92.14

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.9	-4.3	-8.5	2.0	-1.3	-6.9	2.3	-0.4	13.4	9.4	11.9	8.7	0.7	70	0.522

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	34.3	18.6	32.9	18.2	31.1	17.5	19.9	31.4	19.0	30.2	18.3	29.0	3.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.1	12.6	22.3	15.2	11.9	21.5	14.6	11.4	20.9	60.2	31.2	57.4	30.2	54.9	29.1	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.9	9.3	8.0	-8.5	37.3	1.4	1.8	-9.5	38.6	-10.3	39.6	-11.1	40.6	-12.1	41.9		
			-8.8	21.9	1.5	1.5	-9.9	23.0	-10.8	23.9	-11.6	24.8	-12.7	25.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.4	4.4	5.5	8.3	10.7	14.6	19.2	21.4	21.1	17.5	13.2	7.5	4.8
	DBStd	6.86	3.23	3.03	3.00	3.20	3.54	3.45	2.73	2.52	2.90	3.10	3.39	3.30
	HDD10.0	663	174	127	67	30	4	0	0	0	0	8	89	162
	HDD18.3	2449	431	359	310	230	123	32	7	6	48	161	325	418
	CDD10.0	1541	1	2	16	50	148	276	353	346	225	108	14	2
	CDD18.3	285	0	0	0	0	8	57	102	93	23	2	0	0
	CDH23.3	4643	0	0	1	26	211	928	1549	1364	487	76	0	0
	CDH26.7	1983	0	0	0	1	55	389	739	629	156	13	0	0
Wind	WSAvg	2.9	3.1	3.3	3.6	3.6	3.2	2.9	2.5	2.4	2.3	2.5	3.0	2.8
Precipitation	PrecAvg	420	42	31	30	43	45	24	11	12	30	64	47	40
	PrecMax	630	125	79	110	113	114	61	37	33	75	150	198	112
	PrecMin	274	1	2	0	8	6	1	1	0	1	13	4	6
	PrecStd	81	30	22	23	25	27	17	10	11	19	38	37	32
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.7	18.4	22.3	25.9	31.0	35.4	36.2	36.1	33.3	28.4	20.1	15.3
		MCWB	9.5	9.4	10.8	14.1	16.9	19.2	19.0	19.3	18.3	14.8	12.9	10.4
	2%	DB	12.8	15.9	20.3	23.8	28.1	33.0	34.3	34.2	30.2	25.5	17.1	13.3
		MCWB	9.3	9.0	10.2	13.0	15.8	18.4	18.5	18.6	16.7	14.1	11.5	9.9
	5%	DB	11.2	13.8	18.2	21.2	25.7	30.9	32.8	32.4	28.2	22.9	15.0	11.9
		MCWB	8.6	8.6	9.8	12.0	14.7	17.4	18.2	18.0	16.0	13.7	10.9	9.3
	10%	DB	9.9	11.8	15.9	18.6	23.1	28.9	31.0	30.3	26.3	20.5	13.3	10.5
		MCWB	7.8	7.9	9.3	11.2	13.8	16.7	17.7	17.3	15.5	13.3	10.3	8.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.4	11.4	12.8	15.2	18.4	20.5	20.9	21.0	19.4	17.1	14.2	12.0
		MCDB	13.0	14.8	18.5	23.4	28.2	31.5	32.7	33.2	30.5	23.0	17.6	13.6
	2%	WB	10.1	10.2	11.6	13.8	16.6	19.2	19.7	19.7	17.9	15.7	12.8	10.7
		MCDB	11.9	14.0	17.2	21.9	25.8	30.3	31.5	31.1	27.2	21.4	15.6	12.6
	5%	WB	9.1	9.2	10.7	12.7	15.5	18.2	19.0	18.8	17.0	14.9	11.8	9.7
		MCDB	10.8	12.8	16.2	19.7	23.5	29.0	30.5	29.7	25.7	19.9	14.1	11.6
	10%	WB	8.0	8.2	9.7	11.5	14.4	17.3	18.2	18.0	16.2	14.1	10.8	8.6
		MCDB	9.7	11.4	14.7	17.4	21.6	27.1	29.1	28.6	24.3	18.9	13.0	10.4
Mean Daily Temperature Range	5% DB	MDBR	9.3	11.5	13.0	12.5	13.3	15.9	17.5	17.4	15.7	13.3	10.3	9.7
		MCDBR	11.0	15.8	18.5	18.0	18.2	19.6	20.4	20.9	19.8	18.5	13.4	11.0
		MCWBR	7.8	9.6	9.9	8.7	7.6	7.5	7.4	7.6	8.0	8.7	8.2	7.7
	5% WB	MCDBR	9.0	13.1	14.5	15.5	15.8	18.0	19.1	18.8	17.4	13.3	10.3	9.2
		MCWBR	6.8	8.5	8.3	7.9	7.1	7.2	7.3	7.4	7.5	7.0	6.9	6.9
Clear-Sky Solar Irradiance	taub	0.281	0.293	0.326	0.334	0.345	0.349	0.334	0.342	0.329	0.319	0.294	0.279	
	taud	2.515	2.483	2.410	2.398	2.394	2.397	2.409	2.385	2.439	2.486	2.516	2.522	
	Ebn at Noon	890	927	926	938	929	923	934	918	910	883	864	862	
	Edh at Noon	72	87	105	114	117	117	115	114	101	86	72	67	
All-Sky Solar Radiation	RadAvg	1.96	2.98	4.18	5.36	6.50	7.43	7.74	6.73	5.16	3.40	2.20	1.76	
	RadStd	0.24	0.34	0.48	0.55	0.60	0.34	0.27	0.29	0.28	0.33	0.29	0.19	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	+0.95	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	+0.65	N/A	N/A	N/A	N/A	N/A	+57

Nomenclature: See separate page